$n_2 = 5$, $\bar{x}_2 = 8.6$, $s_2^2 = 10.3$

$$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} \approx 12.457$$

$df = 4+5-2 = 7$

$$t^* = \frac{(\bar{x}_1-\bar{x}_2)-(\mu_1-\mu_2)}{\sqrt{s^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \approx -0.675$$

$t_{7,0.025} = 2.365$

$-0.675 > -2.365 \Rightarrow$ not reject.

$t_{7;0.1} > t$

$\Rightarrow 0.1 < p$

$\Rightarrow 0.2 < 2p$

$\Rightarrow 0.05 < 0.2 < P$ value

12. Books or iPads? An experiment was conducted to compare the use of iPads versus regular textbooks in teaching algebra to two classes of middle school students.[6] To remove teacher-to-teacher variation, the same teacher taught both classes, and all teaching materials were provided by the same author and publisher. Suppose that after 1 month, 10 students were selected from each class and their scores on an algebra advancement test recorded. The summarized data follow.

	iPad	Textbook
Mean	86.4 $\bar{x}_1$	79.7 $\bar{x}_2$
Standard Deviation	8.95	10.7
Sample Size	10	10

a. Use the summary data to test for a significant difference in advancement scores for the two groups using $\alpha = .05$.

b. Find a 95% confidence interval for the difference in mean scores for the two groups.

c. In light of parts a and b, what can we say about using iPads versus traditional textbooks in teaching algebra at the middle school level?

c. There is no significant difference

and 95% CI includes 0, which means that it is possible that there's no difference in performance.

a: $H_0: \mu_1-\mu_2=0$, $H_a: \mu_1-\mu_2 \neq 0$

$$t^* = \frac{(\bar{x}_1-\bar{x}_2)-0}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} = \frac{86.4-79.7}{\sqrt{\frac{8.95^2}{10}+\frac{10.7^2}{10}}} \approx 1.52$$

$t_{9,0.025} = 2.262$ $\quad 1.52 < 2.262 \Rightarrow$ not reject

b: $CI: (\bar{x}_1-\bar{x}_2) \pm t_{9;0.025} \cdot \sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}} = 6.7 \pm \underbrace{2.262 \cdot 4.41}_{}$ $= 9.98$

$CI: (-3.28, 16.68)$